Amaury Gouverneur

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Profile

PhD and RL researcher, with a strong background in probability, information theory, and mathematics. My research focuses on decision problems under uncertainty and establishing performance guarantees for learning algorithms, studying how efficiently agents acquire and exploit information in bandits and RL settings. My work combines theoretical analysis with empirical experimentation. Published and presented work at top-tier machine learning journals and venues (ICML, ISIT, NeurIPS, TMLR).

Research Expertise

Reinforcement learning; sequential decision-making under uncertainty; Thompson Sampling; multi-armed and contextual bandits; actor-critic methods; information theory; regret analysis; statistical learning.

Education

Ph.D. in Machine Learning *GPA: 4.0/4.0*
KTH Royal Institute of Technology, Stockholm, Sweden *Aug 2020 – Dec 2025*
Information Science and Engineering (ISE), Electrical Engineering Department

- Thesis title: An Information-Theoretic Approach to Bandits and Reinforcement Learning
- Advisors: Mikael Skoglund & Tobias Oechtering
- Committee: Gergely Neu, Alexandre Proutière, Thomas Schön, Emilie Kaufmann

Visiting Student Researcher
Stanford University, Stanford, CA *Jan 2024 – Jun 2024*
Information Systems Laboratory (ISL), Electrical Engineering Department
Research visit with Prof. Benjamin Van Roy

M.Sc. in Applied and Computational Mathematics *GPA: 4.9/5.0*
KTH Royal Institute of Technology, Stockholm, Sweden *Sep 2018 – Jun 2020*
Minor: Computational Mathematics

M.Sc. in Mathematical Engineering *Summa cum laude: 17.38/20*
École Polytechnique de Louvain, Louvain, Belgium *Sep 2015 – Jun 2020*
Minor: Mathematics of Data Science and Machine Learning
B.Sc. in Electrical and Mathematical Engineering

Work Experience

KTH Royal Institute of Technology *Stockholm, Sweden*
Researcher *Feb 2026 – Present*

- Completing research outputs on generalized linear bandits with frequentist guarantees.
- Developing epistemic actor-critic methods with information-directed objectives.

Lynx Asset Management *Stockholm, Sweden*
Quantitative Research Intern *March – Jun 2025*

- Developed reinforcement-learning methods for trade execution using high-frequency LOB data.
- Implemented actor-critic models for order placement and trained them under realistic market constraints.
- Built simulation and evaluation models to assess execution cost and risk across market conditions.

aSmartWorld *Genval, Belgium*
Founding Engineer & Technical Lead *2019 – 2021*

- Designed technical architecture; developed iOS app for hardware testing and device collection.
- Contributed to fundraising €400k in seed funding; hired and led engineering hires.

Deloitte UK *London, UK*
Strategy & Operations Intern *Summer 2017*

- Quantitative industry and strategy analysis within the Strategy & Operations practice.

Selected Publications (Google Scholar)

- **An Information-Theoretic Analysis of TS for Logistic Bandit Problems** *TMLR 2026*
  Resolves a conjecture on logistic bandits by removing exponential dependence on logistic slope.
- **Refined PAC-Bayes Bounds for Offline Bandits** *ISIT 2025*
  Derives state-of-the-art PAC-Bayes bounds for offline bandits via new optimization technique.
- **Chained IT bounds and Tight Regret Rate for Linear Bandit Problems** *ICML 2024 - FoRLaC*
  Achieves optimal regret rate for linear bandits using a chaining-based analysis.
- **TS Regret Bounds for Contextual Bandits with sub-Gaussian rewards** *ISIT 2023*
  Extends the Russo-Van Roy information-theoretic analysis to contextual bandit problems.

Selected Projects

- Chess-GPT: Fine-tuned and post-trained a large language model to play chess; supervisor of bachelor student thesis projects; demo available at amaurygouverneur.github.io/chess-gpt/.
- Tumor Tracking: Proposed an approximate optimization framework for selecting measurement times in particle-filter-based tumor tracking with limited sensing budgets.

Teaching Experience

- **Pattern Recognition and Machine Learning** *KTH*
  Master's course on Bayesian pattern recognition, HMMs, EM, and variational Bayes. *2020 – 2024*
- **Deep Neural Networks and Generative Models** *KTH*
  Industry course on modern deep learning, CNNs, generative models, and transfer learning. *2022*
- **Machine Learning and Data Science** *KTH*
  Master's course covering Bayesian graphical models, VI, sparse learning, and deep neural networks. *2024*

Programming Skills

Languages: Python, C++, SQL, `MATLAB`
Libraries: PyTorch, JAX, NumPy, Pandas, SciPy, scikit-learn, TensorFlow
Tools & Environment: Linux, Bash, Git, Docker

Service

- Reviewer for ICML, ICLR, ISIT, and ICASSP.
- Research coordination within WASP national research programs for RL and Foundations of AI.

Languages

French (native speaker), English (fluent), Swedish (fluent), German (C1)

Other

Competitive running, math Olympiads, gravel biking, specialty coffee enthusiast